

L2 Norm Bimodality in CALE Clue Embeddings

Overview

When we embed cryptic clue surfaces using CALE’s $\langle t \rangle \langle /t \rangle$ delimiters (f_{clue}), the L2 norms of the resulting 1024-dimensional vectors are visibly bimodal, with two peaks at approximately 29.6 and 31.6 (Figure 1, top row). This report characterizes the bimodality, tests whether it propagates to our cosine-based research metrics, and documents its behavior under fine-tuning.

The Bimodality Is Real and Specific to Clue Context

We applied a two-criterion bimodality test— $\Delta\text{BIC} > 10$ (Kass & Raftery 1995) *and* Ashman’s $D > 2$ (cleanly separated components)—to seven g_{stock} embedding populations. ΔBIC alone flags any departure from normality at large N ; Ashman’s D provides the sample-size-independent effect-size filter.¹

Population	N	ΔBIC	Ashman’s D	Bimodal?
f_{clue} (full)	239,406	28,806	2.88	Yes
f_{clue} (val only)	47,933	5,726	2.88	Yes
f_{wndef} (full)	53,930	400	1.88	No
f_{wnex} (full)	8,360	388	2.11	Yes
$f_{\text{wndef, wnex words}}$	8,360	140	2.12	Yes

The f_{clue} distribution passes both criteria decisively ($D = 2.88$). The f_{wndef} vocabulary embeddings—the same words, decontextualized—fail ($D = 1.88$; Figure 1, middle row), confirming that the bimodality arises from clue context, not from the words themselves.

¹ $D = \sqrt{2}|\mu_1 - \mu_2|/\sqrt{\sigma_1^2 + \sigma_2^2}$; $D > 2$ corresponds to $< 13\%$ overlap between the two Gaussian components.

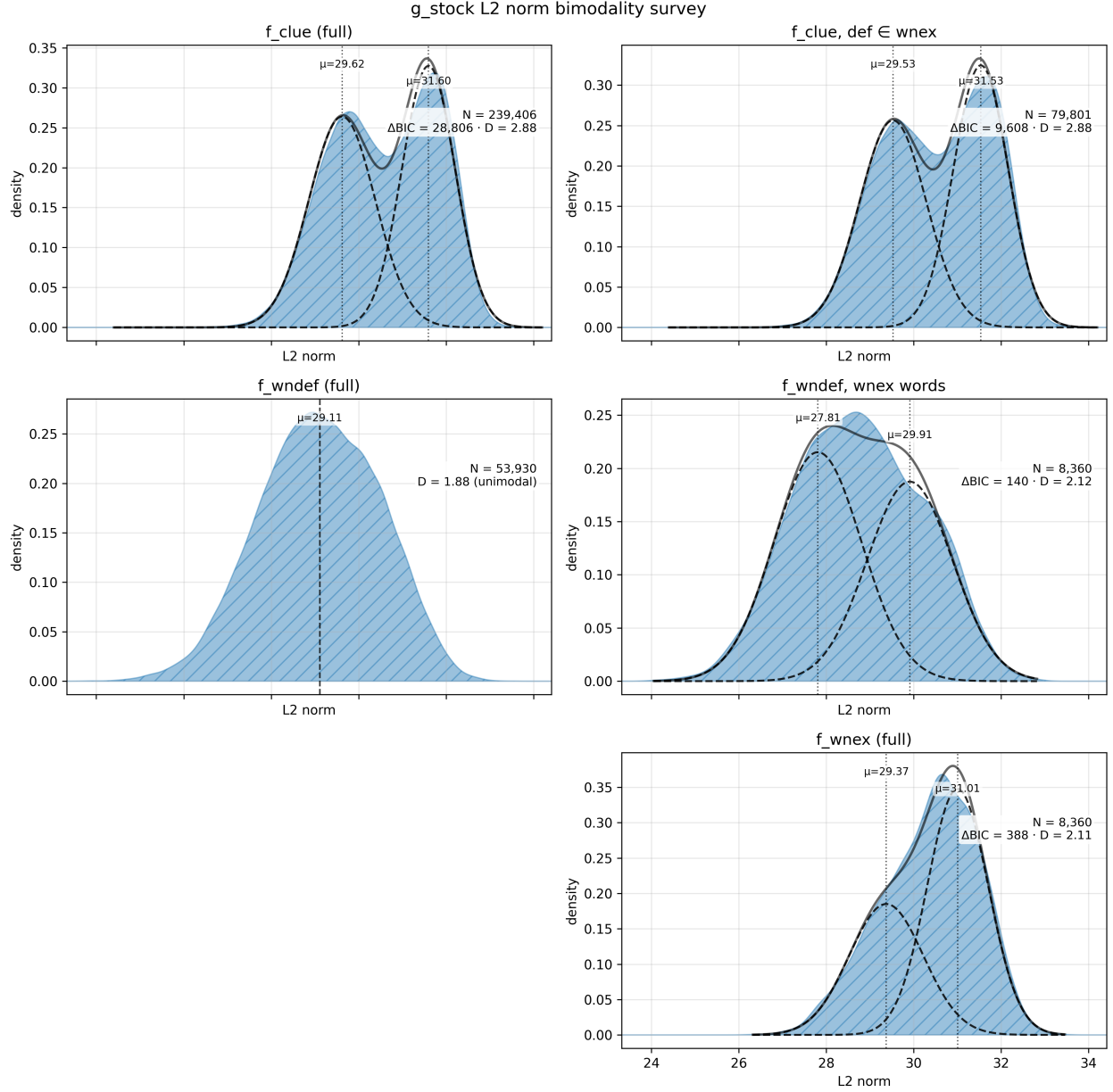


Figure 1: L2 norm distributions for g_{stock} embeddings across five populations. Bimodal populations receive 2-component GMM overlays with annotated component means; unimodal populations show a single mean line. Left column: full datasets; right column: restricted to the 8,360 words in the wnex vocabulary, isolating phrase construction from vocabulary composition. The f_{clue} bimodality (top) persists even on the wnex-restricted subset, while f_{wndef} for the same words (middle right) shows a weaker, less-separated split ($D = 2.12$ vs. 2.88).

What Modulates the Bimodality

Definition position. Whether the tagged definition word appears at the start or end of the clue surface does not create or destroy the bimodality—both subsets show two modes at the same locations ($\mu_1 \approx 29.6$, $\mu_2 \approx 31.6$). What changes is the *balance*: start-of-surface definitions weight the upper mode more heavily ($\pi_{upper} = 0.57$), while end-of-surface definitions weight the lower mode

($\pi_{\text{lower}} = 0.59$). The overall distribution looks bimodal because these two subsets superimpose with different mixture weights.

Not the word, not surface features. An intraclass correlation ($\text{ICC}(1,1) = 0.14$) on per-word f_{clue} norms confirms that most variance is within-word, not between-word: the same definition word receives different norms in different clues. Cross-format Spearman correlations between mean f_{clue} norm and f_{wndef} norm ($\rho = 0.26$) or f_{wnex} norm ($\rho = 0.21$) are weak. A 15-feature regression on surface properties (token counts, definition position, punctuation, nine wordplay-type flags) achieves $R^2 = 0.041$ —96% of the variance is unexplained by any observable text property we tested. The bimodality reflects something about how CALE’s attention mechanism interacts with clue context at a level deeper than surface tokenization.

Magnitude Only, Not Direction

The bimodality is strictly a magnitude phenomenon. A pairwise cosine analysis on 50,000 pairs of unit-normalized f_{clue} embeddings produces a unimodal distribution (mean cosine = 0.318). Splitting embeddings at the median norm and computing the cosine between the two halves’ centroid directions gives 0.978—the lower-norm and upper-norm groups point in essentially the same direction.

No Propagation to Cosine Metrics

Since our research metric is cosine similarity, the practical question is whether the norm bimodality confounds the ATE. Stratifying validation clues by norm group shows nearly identical $T=1$ cosine similarity (lower: 0.523, upper: 0.519) and ATE (lower: -0.053 , upper: -0.047). The Spearman correlation between f_{clue} norm and ATE is 0.019. L2-based $T=1$ distances *do* differ across norm groups (28.2 vs. 29.5), as expected—but this does not leak into cosine. The bimodality and misdirection are independent phenomena.

Fine-Tuning Eliminates the Bimodality

Under g_1 , global norm compression collapses both modes. The component separation shrinks from $\Delta\mu \approx 2.0$ (g_{stock}) to $\Delta\mu \approx 0.9$ (g_1), and the overall standard deviation drops from 1.20 to 0.74. The position-dependent mixture-weight pattern disappears. This is consistent with the general compression documented in the g_1 evaluation: fine-tuning contracted the embedding space uniformly, erasing the magnitude structure that g_{stock} maintained.

Implications

The f_{clue} norm bimodality is a genuine feature of how g_{stock} represents clue-contextualized words—contextual, not lexical, and driven by factors below the surface-feature level. It does not confound cosine-based analyses: ATE measurements are unaffected. L2-based analyses of g_{stock} f_{clue} embeddings should be interpreted with the bimodality in mind, though in practice we report cosine throughout. Under g_1 (or any future fine-tuned model exhibiting similar compression), the concern does not arise.